

≡ ChatGPT ▾

↑ ...

Chunks , Embedding , Vector DB (chroma DB) , Retrieval
Above 4 are the important points used to create a RAG,
now tell me how can we implement these steps and
provide tools and tech stacks used for implementing them.
Remember provide in simple steps as a fresher point of
view.

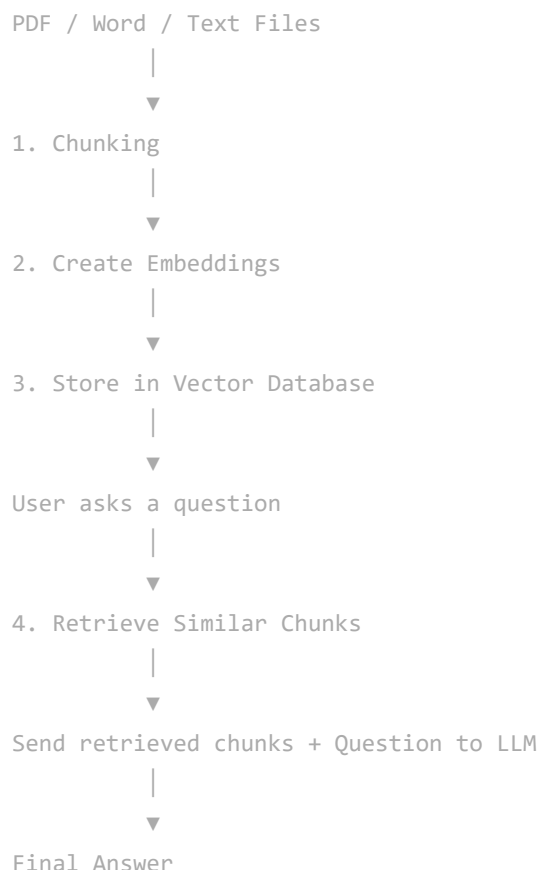


You're correct. These are the **4 core building blocks of a RAG (Retrieval-Augmented Generation) system**.

As a fresher, don't think of RAG as something complex. Think of it like this:

"Instead of asking ChatGPT to answer from its own knowledge, we first search our own documents, then give the relevant information to the LLM so it can answer accurately."

Overall RAG Pipeline



Step 1 : Chunking

What is Chunking?

Suppose your PDF contains **100 pages**.

If you send all 100 pages to the LLM,

- too expensive
- too many tokens
- slower
- context limit exceeded

Instead, we divide the document into **small pieces**.

These pieces are called **chunks**.

Example

Original text

```
Java is an object-oriented programming language.
```

```
It supports inheritance.
```

```
It also supports polymorphism.
```

```
Java applications run on JVM.
```

After chunking

Chunk 1

```
Java is an object-oriented programming language.
```

Chunk 2

```
It supports inheritance.
```

Chunk 3

```
It also supports polymorphism.
```

Chunk 4

Java applications run on JVM.

Which tools do we use?

Most popular:

LangChain

Python

Run

```
RecursiveCharacterTextSplitter
```

Example

Python

Run

```
from langchain.text_splitter import RecursiveCharacterTextSplitter

splitter = RecursiveCharacterTextSplitter(
    chunk_size=500,
    chunk_overlap=50
)

chunks = splitter.split_text(text)
```

Other options

- LlamaIndex
- Haystack
- Custom Python code

Step 2 : Embedding

Now suppose one chunk is

Java supports inheritance.

Computers don't understand English.

They understand numbers.

So we convert this sentence into numbers.

Example

Java supports inheritance

↓

[0.23, -0.67, 0.91, 0.12, ...]

This list of numbers is called an **embedding vector**.

These numbers capture the **meaning** of the text.

Example

Sentence A

Dog is running.

Sentence B

A puppy is running.

Their embeddings will be very close because they have similar meanings.

Sentence C

The stock market crashed.

Its embedding will be far away.

Tools for Embeddings

OpenAI

Python

Run

text-embedding-3-small

Google

Gemini Embedding

HuggingFace

sentence-transformers

Example

all-MiniLM-L6-v2

Very popular for beginners.

Python example

Python

Run

```
from sentence_transformers import SentenceTransformer

model = SentenceTransformer("all-MiniLM-L6-v2")

embeddings = model.encode(chunks)
```

Step 3 : Store in Vector Database

Now suppose we have

Chunk

Java supports inheritance.

Embedding

[0.32,0.67,-0.21,...]

We store both.

Chunk

↓

Embedding

↓

Vector Database

Unlike SQL databases, vector databases are designed to efficiently search based on **similarity** rather than exact matches.

Popular Vector Databases

Vector DB	Beginner Friendly	Free
ChromaDB	 Yes	 Yes
FAISS	 Yes	 Yes
Pinecone		Free tier
Weaviate	Medium	Free tier
Milvus	Medium	Yes
Qdrant	Yes	Yes

For beginners

ChromaDB is one of the easiest choices.

Example

Python

```
from langchain_chroma import Chroma

db = Chroma.from_documents(
    documents=docs,
    embedding=embedding_model
)
```

Run

Now every chunk is stored with its embedding.

Step 4 : Retrieval

Now the user asks

Does Java support inheritance?

The question is also converted into an embedding.

Question

↓

Embedding

↓

Search Vector Database

The vector database compares the question embedding with stored chunk embeddings using similarity measures (commonly cosine similarity).

Suppose it finds

Chunk 17

Java supports inheritance.

That chunk is retrieved.

Usually the top 3–5 most similar chunks are returned.

Example

Retrieved Chunk 1

Java supports inheritance.

Retrieved Chunk 2

Inheritance enables code reuse.

Retrieved Chunk 3

Java also supports polymorphism.

These chunks are then added to the prompt.

Context

Java supports inheritance.

Inheritance enables code reuse.

Question

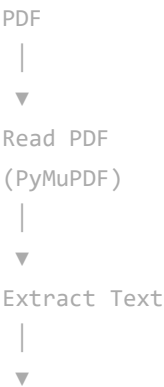
Does Java support inheritance?

The LLM now has the relevant information and generates the answer.

Tech Stack for a Beginner RAG Project

Step	Tool
Programming Language	Python
Read PDF	PyPDF2 or PyMuPDF
Chunking	LangChain RecursiveCharacterTextSplitter
Embedding Model	all-MiniLM-L6-v2 (Sentence Transformers) or OpenAI Embeddings
Vector Database	ChromaDB
Retrieval	LangChain Retriever
LLM	OpenAI GPT, Google Gemini, or another compatible LLM
Frontend (Optional)	Streamlit or React
Backend	FastAPI or Flask

Complete Beginner Workflow




```
Chunk Text
(LangChain)
|
▼
Generate Embeddings
(Sentence Transformers)
|
▼
Store in ChromaDB
|
▼
User Question
|
▼
Convert Question to Embedding
|
▼
Search ChromaDB
|
▼
Top Similar Chunks
|
▼
Send Chunks + Question to LLM
|
▼
Final Answer
```

Interview-Friendly One-Line Explanation

If an interviewer asks "How does RAG work?", you can answer:

"First, we extract text from documents and split it into smaller chunks. Then we convert each chunk into embeddings using an embedding model and store those embeddings in a vector database like ChromaDB. When a user asks a question, we convert the query into an embedding, retrieve the most semantically similar chunks from the vector database, and provide those chunks as context to the LLM, which uses them to generate an accurate, context-aware response."



Do you like this personality?



